LAB 7 — FUZZ TESTING FOR INPUT VALIDATION BUGS

Gym Membership Application | Student: Shamika

1. Objective

Apply Hypothesis property-based fuzz testing to the assigned gym membership application, discover edge-case input failures, and harden the application through defensive input validation.

2. Application under test

The repository's Lab 7 application contains three functions: **parse_plan** (plan code and duration), **calculate_bmi** (weight and height), and **average_workout** (workout-minute average). The repository's actual filenames are retained instead of the handout's generic processor.py names.

3. Baseline and fuzzing results

The original application was run with GOLD-12, 70,1.75, and 45 60 30. It returned ('GOLD', 12), BMI 22.9, and average workout minutes 45. Hypothesis was then run against the buggy module and reported **4 failed and 3 passed**. One minimized failure was None passed to parse_plan, causing an AttributeError because None has no split method.

4. Fixes implemented

parse_plan now validates input type, exact field count, alphabetic plan name, numeric month count, and positive duration. calculate_bmi validates format, numeric and finite values, and positive weight and height. average_workout rejects non-string, empty, malformed, and negative-minute input. Invalid values return None instead of causing uncaught exceptions.

5. Final verification

After switching the tests to the fixed gym_membership module, the student ran `python3 -m pytest -v test_gym_membership.py` on macOS with Python 3.13.7, pytest 9.2.0, and Hypothesis 6.170.15. The captured result was **7 passed in 0.38s**.

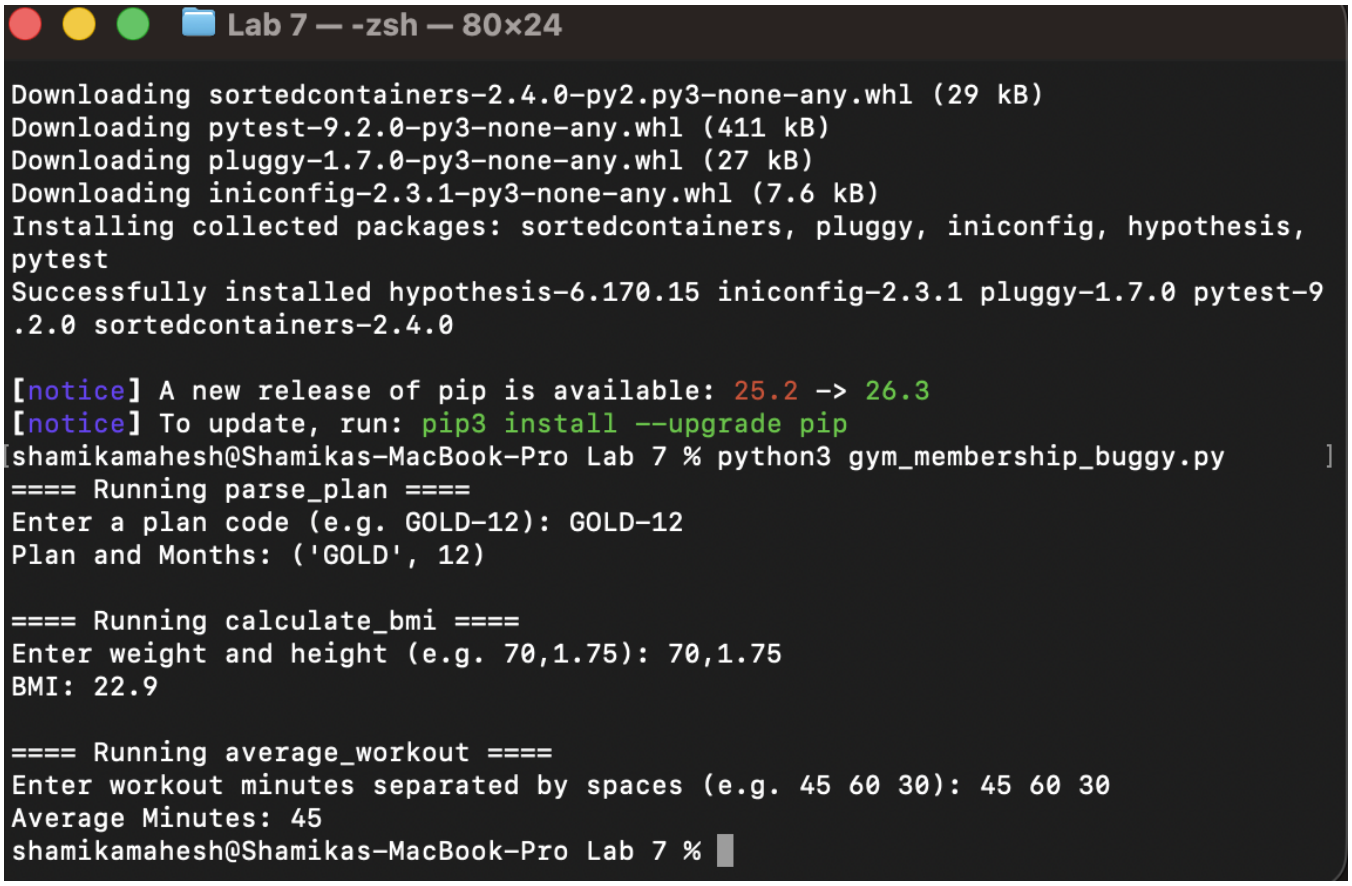
6. Conclusion

Hypothesis exposed edge cases beyond ordinary valid-input demonstrations. Defensive validation was added to all three functions, and the final seven-test suite passed.

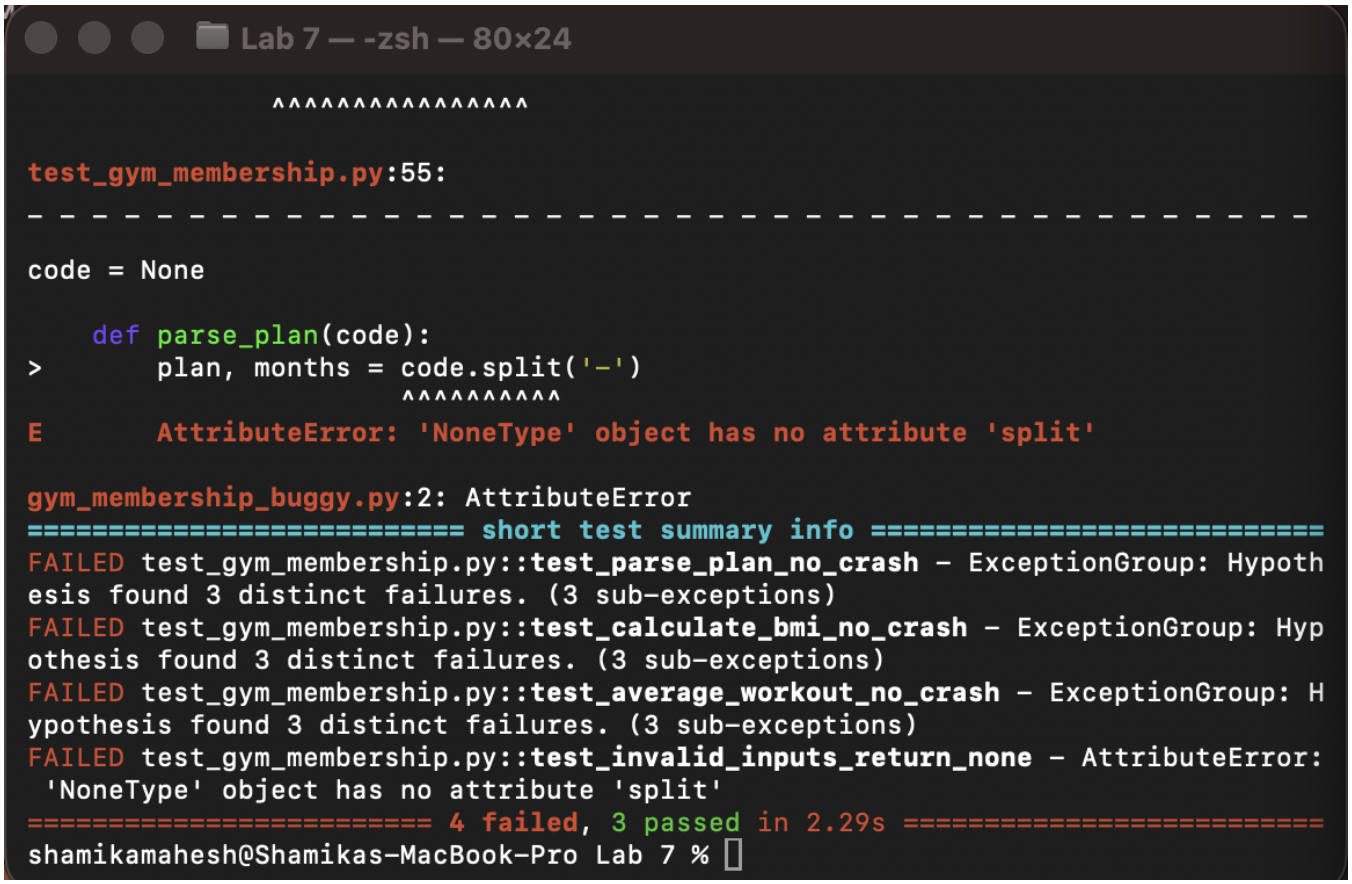
Execution Evidence

The following screenshots were captured during the student's actual terminal run.

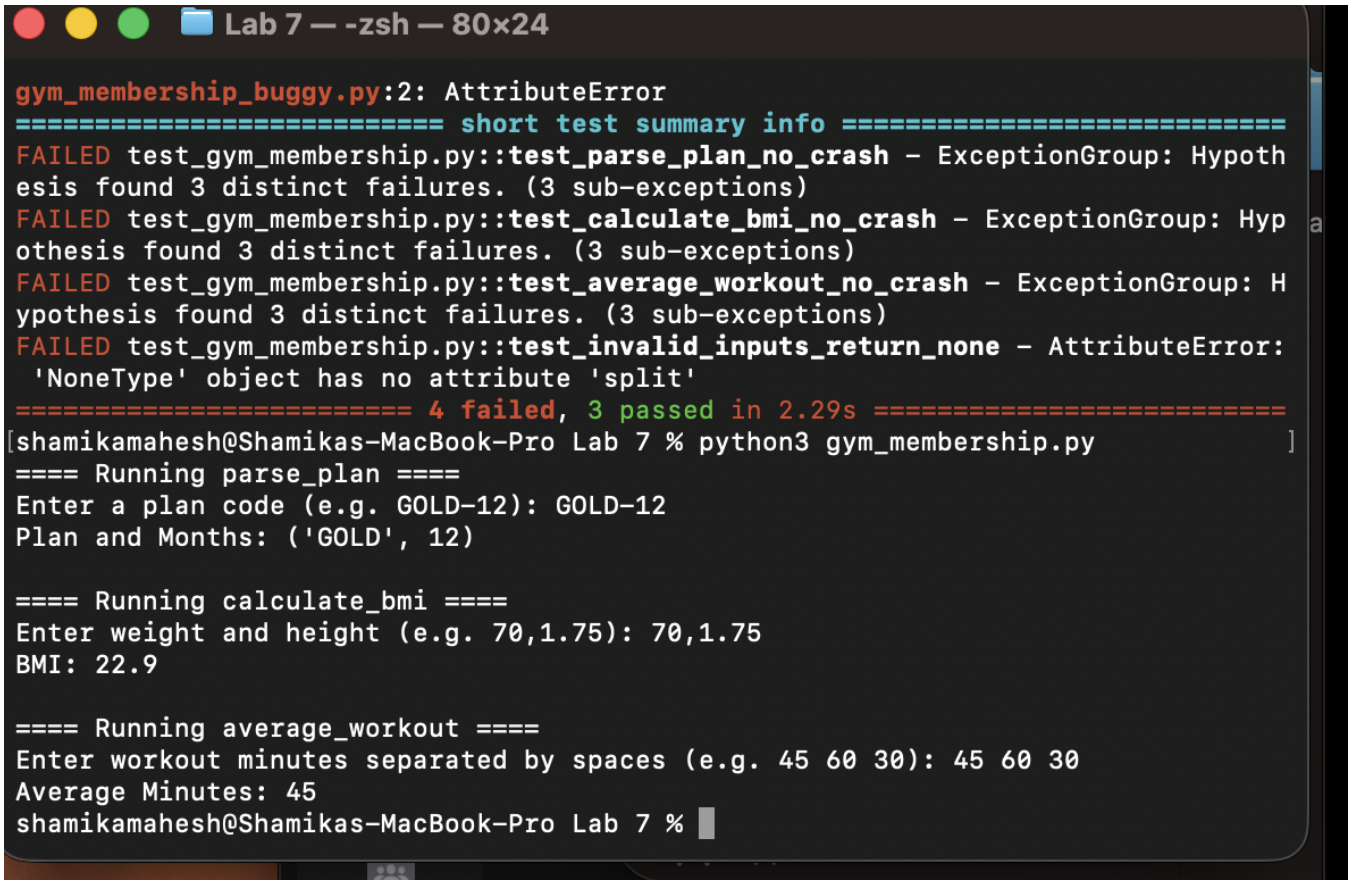
Screenshot 1 — Original application executed with valid sample inputs.



Screenshot 2 — Hypothesis against buggy code: 4 failed, 3 passed.



Screenshot 3 — Fixed application executed with the same valid sample inputs.



Screenshot 4 — Final Hypothesis test run: 7 passed in 0.38 seconds.

